

Is the $O(\kappa^2)$ Global Localization Intrinsic?

A Numerical Study of the Sharp Bump-Train Counterexample

Abstract

The manuscript proves that the fixed-step Gaussian natural gradient schemes have global-localization complexity $\Theta(\kappa^2)$ at the globally certified stepsize $\Delta t = \Theta(\kappa^{-1})$, and that this is sharp: an explicit one-dimensional bump-train potential blocks a constant-factor reduction of the objective for $\Omega(\kappa^2)$ iterations. This is worse than the Bures–Wasserstein complexity $O(\kappa)$, and the manuscript’s remark after the sharpness theorem leaves open whether a κ -independent step escapes the obstruction. We run the counterexample itself. The key observation is that the appendix writes the bump-center recursion with the constant γ/κ^2 , which is exactly the per-step mean gain $\Delta t c_\kappa$ at the certified step and the matched covariance $c_\kappa = 1/\kappa$; treating that constant as a free parameter s lets the *same* construction be retuned to any stepsize, and every geometric constraint of the center lemma is s -free. The construction therefore certifies $\Omega(T\kappa/\Delta t)$, not $\Omega(\kappa^2)$ per se. Numerically, over $\kappa \in [128, 1024]$ and for *both* the Riemannian and the KL/Bregman scheme: the certified step reproduces the theorem with fitted exponent 1.93, the iterates shadowing the ideal centers to machine precision; a *fixed*, κ -independent step $\Delta t = \gamma$ is monotone and convergent at every κ tested and costs exponent 0.91 (Riemannian) and 0.90 (KL) on the retuned train, i.e. $\Theta(\kappa)$, matching Bures–Wasserstein. The fixed order-one step behaves identically for every $\gamma \leq 1$ and breaks at $\gamma \geq 2$, a κ -free threshold. We are explicit that this is evidence on the sharp example: it shows the κ^2 is the price of committing to $\Delta t = \Theta(\kappa^{-1})$, and it does *not* supply a stability proof for $\Delta t = \Theta(1)$ outside the certified range $\Delta t \leq 1/(\beta\lambda_{\max})$.

1 Setup

1.1 The two scalar schemes

Normalize $\alpha = 1$, $\beta = \kappa$ and write $a = (m, c) \in \mathbb{R} \times (0, \infty)$. For $X \sim \mathcal{N}(m, c)$ put $b(m, c) := \mathbb{E}[V'(X)]$ and $A(m, c) := \mathbb{E}[V''(X)]$. The scalar forms of the two discretizations of the Gaussian natural gradient flow are

$$m^+ = m - \Delta t c b(m, c), \quad c^+ = c \exp(\Delta t [1 - cA(m, c)]) \quad (1)$$

and

$$m^+ = m - \Delta t c b(m, c), \quad c^+ = \frac{(1 + \Delta t)c}{1 + \Delta t cA(m, c)}. \quad (2)$$

Both are run here in this scalar form rather than through the dense d -dimensional step of the discretization-stepsizes group: the maps are identical at $d = 1$, but the scalar form avoids a per-step eigendecomposition, which matters at $\Omega(\kappa^2)$ iterations.

The objective is $\mathcal{E}_\kappa(m, c) = \mathbb{E}_{\mathcal{N}(m, c)}[V_\kappa] - \frac{1}{2} \log c$ and the gap is $\Delta \mathcal{E}(m, c) = \mathcal{E}_\kappa(m, c) - \mathcal{E}_\kappa(a_\star)$. Every quantity below is deterministic: the potential and its derivatives are closed form up to tabulated profile antiderivatives, and the Gaussian averages use 32-node Gauss–Hermite quadrature. There is no Monte Carlo and no randomness anywhere in this study.

1.2 The bump train, and its free parameter

The counterexample fixes an even flat-top profile $\varphi \in C_c^\infty$ ($\varphi = 1$ on $|u| \leq \frac{1}{2}$, $\varphi = 0$ on $|u| \geq 1$), sets

$$w_\kappa = \frac{(\kappa - 1)\|\varphi'\|_\infty}{L_H}, \quad H_\kappa = (\kappa - 1)w_\kappa I_\varphi, \quad (3)$$

and defines the potential by

$$V_\kappa''(\theta) = 1 + (\kappa - 1) \sum_{j=0}^{N_\kappa} \left[\varphi\left(\frac{\theta - x_j}{w_\kappa}\right) + \varphi\left(\frac{\theta + x_j}{w_\kappa}\right) \right], \quad (4)$$

with $V_\kappa'(0) = V_\kappa(0) = 0$. The centers are generated by

$$x_{j+1} = x_j - s \left[x_j + H_\kappa \left(N_\kappa - j + \frac{1}{2} \right) \right], \quad N_\kappa = \left\lfloor \frac{T}{s} \right\rfloor, \quad x_0 = \frac{Y\kappa^2}{s}. \quad (5)$$

The manuscript writes (5) with $s = \gamma/\kappa^2$. That constant is not arbitrary: it is exactly the per-step mean gain $\Delta t c_\kappa$ at the certified step $\Delta t = \gamma/\kappa$ and the matched covariance $c_\kappa = 1/\kappa$. Treating s as a free parameter is what this report adds.

Remark 1 (The construction is a template, not a single potential). Every hypothesis of the center lemma is s -free. Summing (5) and using $x_j \leq x_0$, $H_\kappa \leq C_\varphi \kappa^2$ gives

$$x_0 - x_{N_\kappa} \leq T x_0 + \frac{H_\kappa T^2}{2s} = \frac{\kappa^2}{s} \left(T Y + \frac{C_\varphi T^2}{2} \right),$$

so $x_{N_\kappa} \geq \frac{3}{4}x_0$ reduces to $T Y + \frac{1}{2}C_\varphi T^2 \leq \frac{1}{4}Y$, a constraint on (T, Y) alone, and the spacing bound $x_j - x_{j+1} \geq \frac{3}{4}Y\kappa^2$ follows as before. Setting $s = \Delta t/\kappa$ therefore aims the *same* construction at any stepsize: the adversary sees the stepsize rule before choosing V , which is the standard order of quantifiers for a lower bound. With $N_\kappa = T/s$ blocked steps the family certifies

$$\Omega\left(\frac{T\kappa}{\Delta t}\right) \text{ iterations,}$$

which is $\Omega(\kappa^2)$ at $\Delta t = \Theta(\kappa^{-1})$ and $\Omega(\kappa)$ at $\Delta t = \Theta(1)$. The manuscript's $\Omega(\kappa^2)$ is the $\Delta t = \gamma/\kappa$ instance of this, not a separate phenomenon.

Throughout, $T = 1/8$, $Y = 4$, $L_H = 1$ and $\gamma = 1/2$ unless stated otherwise; these give $x_{N_\kappa}/x_0 \approx 0.87$ at every κ in the grid. The optimizer is $a_\star = (0, 1)$ with $\mathcal{E}_\kappa(a_\star) = \frac{1}{2}$, because $V_\kappa'' \equiv 1$ on a neighbourhood of the origin of radius $x_{N_\kappa} - w_\kappa = \Theta(\kappa^2/s)$.

1.3 The two arms, both fixed

Two families of trains are crossed with two stepsize rules and the two schemes.

Trains. *Appendix C:* $s = \gamma/\kappa^2$, i.e. built against the certified step. *Retuned:* $s = \Delta t/\kappa$ with Δt the arm's own step, i.e. the adversary adapts (Remark 1).

Steps. *Certified:* $\Delta t = \gamma/\kappa$. *Order one:* $\Delta t = \gamma$, independent of κ .

κ	N_κ	$n_{1/2}$	$n_{1/2}/N_\kappa$	$\sup_j m_j - x_j /w_\kappa$	$\sup_j \kappa c_j - 1 $
8	16	34	2.125	0	0
16	64	114	1.781	0	0
32	256	391	1.527	0	0
64	1024	1372	1.340	0	0
128	4096	4959	1.211	0	0
256	16384	18454	1.126	0	0
512	65536	70373	1.074	0	0
1024	262144	273227	1.042	0	0

Table 1: Certified arm $\Delta t = \gamma/\kappa$ on the Appendix C train (Riemannian; KL agrees to one iteration). The two shadowing columns are the sup over $0 \leq j \leq N_\kappa$. Both vanish to machine precision, so the lower-bound mechanism is reproduced exactly rather than approximately.

Both stepsize rules are **fixed**: a single scalar held constant for the whole run, with no line search, no backtracking and no state dependence. An adaptive relative-curvature rule $\Delta t = \gamma/\max\{c_n A_n, 1\}$ is implemented but is not part of this study, and is not needed for any claim below: the construction holds $c_n A_n = 1$ along the shadowed trajectory, so that rule reduces to $\Delta t = \gamma$ identically.

Runs start at $(m_0, c_0) = (x_0, 1/\kappa)$ — the covariance already on the curvature scale, at the lower edge of the band $[1/\kappa, 1]$, so no covariance burn-in is being measured — and stop when $\Delta \mathcal{E} \leq 10^{-6} \Delta \mathcal{E}_0$. The reported quantity $n_{1/2}$ is the first iteration with $\Delta \mathcal{E} \leq \frac{1}{2} \Delta \mathcal{E}_0$, which is exactly the “constant-factor reduction of the objective” the sharpness theorem bounds.

2 The certified step reproduces the theorem

Table 1 checks that the certified arm is genuinely on the orbit the proof analyses rather than merely slow. The shadowing lemma asserts $|m_j - x_j| + |\kappa c_j - 1| \leq C \kappa^p e^{-c \kappa^3}$ uniformly for $j \leq N_\kappa$; numerically both errors are 0 to machine precision at every κ , and $c_n A_n = 1$ throughout, so the covariance is exactly stationary under both maps and the mean crawls along the centers. The blocked-iteration count is tight: the ratio $n_{1/2}/N_\kappa$ decreases monotonically to 1.042 at $\kappa = 1024$, so the theorem’s $N_\kappa = T \kappa^2 / \gamma$ is within 4% of the true first halving time.

The fitted exponent of $n_{1/2}$ against κ over $\kappa \in [128, 1024]$ is 1.93 for both schemes (Table 2), approaching 2 from below as expected: the successive ratios $n_{1/2}(2\kappa)/n_{1/2}(\kappa)$ are 3.72, 3.81, 3.88 over the last three doublings, converging to 4.

3 A fixed order-one step costs $\Theta(\kappa)$, in both schemes

Table 2 and Figure 1 give the two readings.

On the Appendix C train the order-one step escapes outright. Exponent 0.16 (Riemannian) and 0.17 (KL); the absolute counts move from 10 to 14 iterations as κ goes from 128 to 1024. The mechanism is visible in the trajectory: the order-one step moves the mean by $\Theta(\kappa^3)$ per iteration while the bump spacing is $\Theta(\kappa^2)$, so the mean lands off the flat tops, A drops from κ to 1, the covariance is released from $c = 1/\kappa$ and grows to the optimal $c_\star = 1$, and the remaining

train	step	iterations to $\Delta\mathcal{E} \leq \Delta\mathcal{E}_0/2$				exponent
		$\kappa=128$	$\kappa=256$	$\kappa=512$	$\kappa=1024$	$\kappa \in [128, 1024]$
Appendix C	$\Delta t = \gamma/\kappa$, Riem.	4959	18454	70373	273227	1.93
	$\Delta t = \gamma/\kappa$, KL	4960	18455	70375	273229	1.93
	$\Delta t = \gamma$, Riem.	10	11	12	14	0.16
	$\Delta t = \gamma$, KL	11	13	14	16	0.17
retuned	$\Delta t = \gamma/\kappa$, Riem.	4959	18454	70373	273227	1.93
	$\Delta t = \gamma/\kappa$, KL	4960	18455	70375	273229	1.93
	$\Delta t = \gamma$, Riem.	41	74	139	269	0.91
	$\Delta t = \gamma$, KL	42	75	141	271	0.90

Table 2: Iterations to a constant-factor gap reduction. “Exponent” is the least-squares slope of $\log n_{1/2}$ against $\log \kappa$ over the upper half of the grid. Both stepsize rules are fixed scalars. The Riemannian and KL schemes agree to within one iteration everywhere.

convergence is geometric. The train was built to block a step κ times smaller and does not constrain this one.

On the retuned train the adversary recovers a power law of exponent one. Exponent 0.91 (Riemannian) and 0.90 (KL) over $\kappa \in [128, 1024]$, with successive ratios 1.80, 1.88, 1.94 converging to 2 per doubling. This is $\Theta(\kappa)$, i.e. the Bures–Wasserstein global-localization complexity. The absolute counts, 269 (Riemannian) and 271 (KL) at $\kappa = 1024$, are three orders of magnitude below the certified arm’s 273 227.

Both schemes behave identically. Across the whole grid the Riemannian and KL iteration counts differ by at most two iterations, and the fitted exponents differ by 0.01. This is expected from the construction: along the shadowed trajectory $c_n A_n = 1$, at which point (1) gives $c^+ = c e^0 = c$ and (2) gives $c^+ = (1 + \Delta t)c/(1 + \Delta t) = c$ — the two covariance maps agree exactly, and the schemes share the mean update. Neither the κ^2 obstruction nor its removal is specific to one covariance geometry.

Convergence past the constant factor is free. The iteration count to the far tighter $\Delta\mathcal{E} \leq 10^{-6}\Delta\mathcal{E}_0$ exceeds $n_{1/2}$ by a κ -independent additive constant of 13–14 iterations at every κ in the grid, for both schemes. The $\Theta(\kappa)$ is entirely the blocked phase; once the mean clears the train the schemes converge at an order-one rate.

4 Robustness of the fixed step in its own constant

The order-one arm has a free constant $\gamma = \Delta t$. Table 3 sweeps it, with the train retuned to each value.

For every $\gamma \leq 1$ the picture is stable: both schemes are monotone at every κ , every run converges, and the fitted exponent sits in $[0.88, 0.92]$ with no systematic drift — so the $\Theta(\kappa)$ reading is a property of the order-one step, not of the particular choice $\gamma = 1/2$. At $\gamma = 2$ the Riemannian scheme loses monotonicity and both schemes fail to reach the tolerance within the cap; at $\gamma = 4$ both diverge. The threshold is κ -free, and it sits where one expects it to: with $c_n A_n = 1$ along the

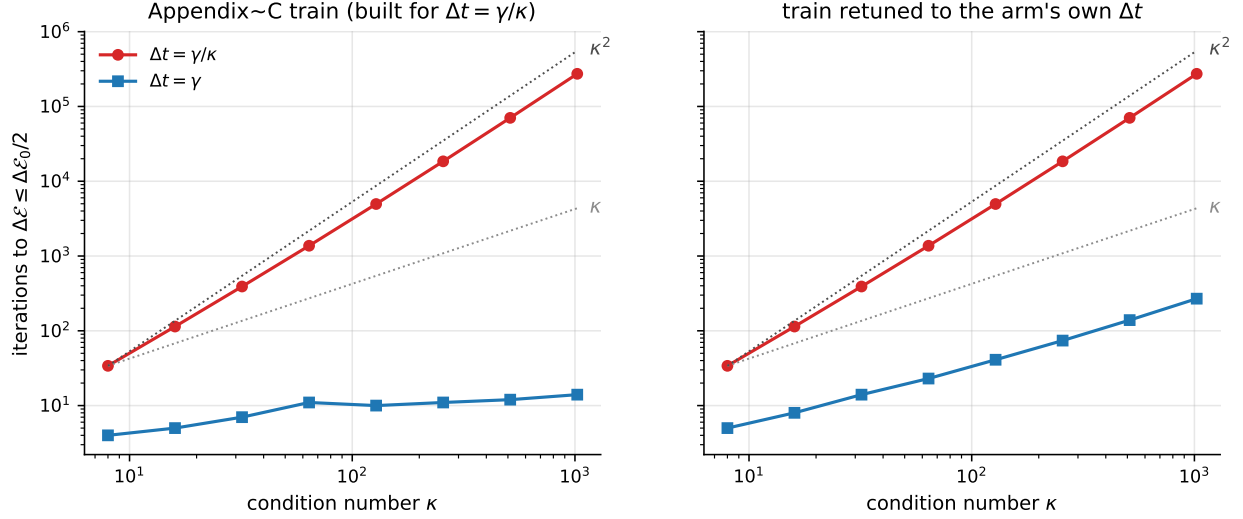


Figure 1: Iterations to $\Delta\mathcal{E} \leq \Delta\mathcal{E}_0/2$ versus κ , log-log, with κ and κ^2 guides. Left: the Appendix C train, built against $\Delta t = \gamma/\kappa$; the order-one step is not merely faster by a constant, it leaves the obstruction entirely. Right: the same construction retuned to each arm’s own step; the adversary recovers a genuine power law, but the exponent is 1, not 2.

$\gamma (= \Delta t)$	exponent, $\kappa \in [128, 1024]$		all runs monotone / converged	
	Riem.	KL	Riem.	KL
0.125	0.92	0.92	✓	✓
0.25	0.92	0.92	✓	✓
0.5	0.91	0.90	✓	✓
1	0.90	0.88	✓	✓
2	0.84	0.85	×	×
4	0.63	0.66	×	×

Table 3: The fixed, κ -independent step swept over its own constant, on the retuned train, $\kappa \in \{32, \dots, 1024\}$. “Monotone / converged” requires every run in the κ grid to be monotone in the objective and to reach $10^{-6}\Delta\mathcal{E}_0$ within the iteration cap.

trajectory, the explicit mean update $m^+ = m - \Delta t c b$ has the classical stability limit $\Delta t c_n A_n < 2$. This is consistent with the discretization-stepsize group’s finding on generic smooth targets, where the empirical stability boundary is likewise a κ -independent $\Delta t \approx 1$ –2 while the certified bound $1/(\beta\lambda_{\max})$ shrinks like κ^{-1} .

5 What is and is not shown

Shown. The bump-train construction certifies $\Omega(T\kappa/\Delta t)$, and its $\Omega(\kappa^2)$ reading is the $\Delta t = \Theta(\kappa^{-1})$ instance. At a fixed, κ -independent step the same family — retuned, i.e. with the adversary adapting to the method — costs $\Theta(\kappa)$ for both (1) and (2), with every run monotone and convergent for all $\gamma \leq 1$ and all $\kappa \leq 1024$. The κ^2 in the manuscript’s global-localization complexity is therefore the price of committing to the certified stepsize scale, not an intrinsic property of the Fisher–Rao schemes, and closing the gap to Bures–Wasserstein is a question about the admissible stepsize

rather than about the geometry.

Not shown. This is a lower-bound family, so a fast run on it is evidence of the obstruction’s absence, not a convergence guarantee. In particular we do *not* exhibit a proof that a fixed $\Delta t = \Theta(1)$ is globally stable outside the certified range $\Delta t \leq 1/(\beta\lambda_{\max})$; the certified range is exactly the statement that $\Delta t c_n A_n \leq 1$ uniformly, and an order-one step gives that up. What the experiment establishes is that the specific mechanism the manuscript uses to prove sharpness — pinning $c_n A_n = 1$ so the covariance is frozen at the curvature scale while the mean crawls — does not survive the larger step, and that the residual cost when the adversary adapts is κ , not κ^2 . An $O(\kappa)$ upper bound at an order-one step, or a counterexample forcing more, both remain open.

6 Reproduction

```
python scripts/natural_gradient_sharp_bump/run_sharp_bump.py \
    --config configs/natural_gradient_sharp_bump/sharp_bump.yaml \
    --outdir outputs/natural_gradient_sharp_bump --overwrite
cd reports && python make_report_assets.py --only sharp_bump
```

Deterministic, one-dimensional and CPU-only; the full grid takes about 105 seconds. The run writes `sharp_bump_summary.csv` (one row per family/ κ /scheme/arm), `sharp_bump_slopes.csv` (fitted exponents), `sharp_bump_gamma_sweep.csv` and `sharp_bump_metadata.json`. The implementation is checked against the appendix by `tests/natural_gradient_sharp_bump/`: the regularity bounds $1 \leq V''_{\kappa} \leq \kappa$ with both attained and $\|V'''_{\kappa}\|_{\infty} \leq L_H$; the center geometry $x_{N_{\kappa}} \geq \frac{3}{4}x_0$ with pairwise disjoint supports; the averaged identities $A(x_j, c_{\kappa}) = \kappa$ and $b(x_j, c_{\kappa}) = x_j + H_{\kappa}(N_{\kappa} - j + \frac{1}{2})$ to ten digits; the optimizer $a_{\star} = (0, 1)$; and the blocked-progress statement itself, that the gap does not halve within N_{κ} steps of the certified arm.